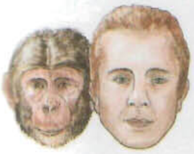
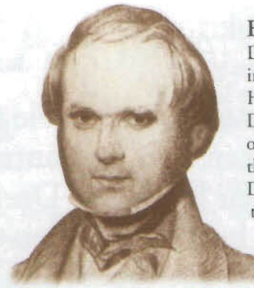
DARWIN, CHARLES



THE BRITISH NATURALIST Charles Darwin is best known as the man who developed the remarkable theory of evolution by natural selection. The theory, which describes how one species can develop or evolve into another, caused a revolution in biological science. Darwin was not the first person to suggest a theory of evolution, but was the first to present a solid body of evidence for the idea. He also wrote books about his travels, coral reefs, barnacles, the pollination of flowers, and insect-eating plants.



Early life
Darwin was born in 1809, in Shrewsbury, England. His grandfather, Erasmus Darwin, had put forward his own theory of evolution in the 1790s. At first, Charles Darwin did not believe in the idea of evolution. He trained as a priest before studying geology and biology.

Galápagos Islands

Darwin studied thousands of plants and animals all around the world on the *Beagle*'s journey. The most interesting part was the few weeks spent in the Galápagos Islands, about 1,000 km (600 miles) from the coast of South America. Darwin noticed that the species there were different from those elsewhere in the world.

Notebooks used by Darwin in Galápagos

List of species

Map pasted into notebook

Galápagos Islands

Notebooks
During his voyage on the *Beagle*, Darwin made careful, copious notes of everything he saw, gaining him the nickname "the old philosopher" from the ship's officers. The wealth of information he gathered helped him later, when he was developing his theory of evolution.

Darwin's finches
When he got home, Darwin realized that the finches on the Galápagos Islands had different beaks, depending on which island they inhabited. He decided that the birds had developed beaks that were best suited to the diet on their particular island.

The Beagle

At Cambridge, Darwin made friends with John Henslow, the professor of botany. Henslow suggested that Darwin would be a good choice as official naturalist on the naval survey ship HMS *Beagle*, which was about to sail around the world on a five-year scientific cruise. The trip lasted from 1831 to 1836.

Darwin's watch

Darwin's telescope

Fossil finds
When he landed in South America, Darwin found fossils of extinct animals, such as the giant sloth (now called *Mylodon darwini*), that closely resembled modern species. This suggested that animals had gradually changed to suit their environments.

Bones of Macrauchenia, *a prehistoric mammal found by Darwin*

Rock hammers

The Origin of Species

Darwin returned to England and wrote an account of his travels. He spent years studying the specimens he had collected and the notes he had made. Gradually, he developed his idea that species evolved as animals adapted to suit their environments. He published his findings in his book *On the Origin of Species by Means of Natural Selection.* The work caused an outcry among Christians because it challenged the creation story in the Bible.

The naturalist
After his voyage, Darwin spent the rest of his life studying specimens, doing experiments, and writing up his findings. He never left England again, and for much of the rest of his life he was too ill to leave his home. Illness did not stop him working on subjects ranging from earthworms to the pollination of plants.

Some of Darwin's equipment

Scissors for dissection

Hand lens

Specimen boxes, one containing butterfly

Seeds sent to Darwin

Slide

Wallace
The British naturalist Alfred Russel Wallace (1823–1913) drew up a theory of evolution by natural selection quite independently of Darwin. He wrote to Darwin for advice, and the men wrote a paper about evolution together.

Natural selection
Parents produce many offspring, all different from each other. Only those best suited to their environment will survive, passing on some of their features to their offspring.

Lesser black-backed gull

Herring gull has same ancestor as Lesser black-backed, but has evolved separately.

CHARLES DARWIN

1809	Born in Shrewsbury, England.
1831	Sets sail on the *Beagle.*
1836	Returns to England.
1858	Wallace writes to Darwin about his evolutionary theory; they produce a paper on evolution together.
1859	Darwin publishes his *Origin of Species.*
1871	Publishes *The Descent of Man*, on human evolution.
1875	Publishes *Insectivorous Plants*, which describes how the sundew traps insects.
1880	Publishes *The Power of Movement in Plants*, which shows how light influences the direction of plant growth.
1882	Dies at Downe, England.

FIND OUT MORE
BIOLOGY · DINOSAURS · EVOLUTION · EXPLORATION · FOSSILS · GEOLOGY · HUMAN EVOLUTION · SCIENCE, HISTORY OF